Supplementary Material: Controlling Creator Exposure under Recommendation Feedback

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Shared CWOP notation follows the main paper; local notation is introduced below. Labels beginning with M refer to the main paper; labels beginning with S refer to this document. Bibliographic numbering is local.

S1. Supplement and Computational Artifact Guide

The accompanying ZIP contains this document and its LaTeX source, together with `artifact/`: experiment runners, controller implementations, tests, and a pinned Python environment. Datasets, saved numerical outputs, graph and policy arrays, run logs, and verification records are not bundled. This document retains theory, experimental protocols, and code instructions; numerical results appear only in the main paper.

a) Code and experiment map: Static synthetic optimization uses `run_static.py`; matched feedback uses `run_unified_safety.py`. The latter generates separate training trials, selected parameters, held-out trajectories, per-window residuals, and certificate diagnostics. The segment-only comparison and fixed-score sample-size curves use `run_safety_refinement.py`; graph-to-graph error bars use `run_sample_size_replicates.py`. The empirical-margin and same-state diagnostics use the review experiment scripts. Public static optimization uses `tune_lastfm.py` and `run_lastfm.py`. The additional public-feed and joint policy-gradient comparisons use `prepare_kuairand.py` and `run_public_feed.py`; `validate_public_feed.py` independently replays their sampled windows and checks margins. Diagnostic urn runners are in `legacy/`. The artifact README gives the dependency order and exact commands.

b) Verification after rerunning: After installing `requirements-lock.txt`, the synthetic unit tests can be run directly. Result validators require the corresponding experiments to be run first. `validate_unified.py` checks matched-window policies, replays selected frozen-policy draws, and reruns a complete certified synthetic trajectory. It also checks training/test separation and training-only baseline selection. `replay_exposure_outcomes.py` regenerates the request, item, and response draws and checks the resulting window summaries. `render_relative.py` reconstructs relative tables, numerical macros, and figures from locally generated results. `compare_matched_policies.py` checks the utility comparison on every held-out graph/split and generates the paired gains and reporting macros. The `-recomputed` commands record checks on local outputs; they do not establish identity with the original experiment files, which are not distributed.

c) Full reruns and environment: The original runs used Python 3.9.6 on macOS arm64 with NumPy 2.0.2, CVXPY 1.7.5, Clarabel 0.11.1, and SCS 3.3.1. Full scripts and seeds are retained; the instructions distinguish commands that resume cached trials from commands that recompute outputs. Solver and hardware changes can perturb numerical trajectories and runtime. Check feasibility residuals and the independent tangent-dual upper/lower bounds; a solver status alone is insufficient. Output directories are created by the runners when needed; empty folders are not included in the ZIP.

d) Data availability: No datasets or saved result files are included. Synthetic inputs are generated by the runners. The official HetRec 2011 Last.fm release must be obtained separately using the checksum-checking download script, subject to its noncommercial-use restriction and upstream notice [1]. The benchmark uses only numeric listening relations. KuaiRand-Pure is obtained separately through `download_kuairand.py`, which checks the publisher checksum [2]. The main paper additionally reports rounded relative platform-audit findings; Section S5.F describes their protocol. Raw company records and internal field names are excluded. The audit provides observational evidence, while the synthetic and public-data tests evaluate CWOP.

e) Protocol separation: The current certified comparison uses 20k synthetic and 100k Last.fm or KuaiRand requests per window, 20 windows, and final-policy-dependent certificates. Historical 10k empirical margin diagnostics and 50k analytic-margin experiments are separately identified in the code and this document; they are not interchangeable with the matched-window results. The code retains the failure checks and fallback behavior used by these protocols.

A. Compact implementation

The compact implementation stores UK eligible probabilities for equal-size candidate sets of size K and sorts each row

for Euclidean simplex projection. It uses the same current-state residuals, pre-update average, and x-coordinate gradients as Algorithm S1. Finite-difference checks and an equivalence test compare it with the original loop implementation. Eligible-edge storage is $O(UK)$, excluding the conic solver and saved traces; this is not a peak-memory benchmark. Static optimizer runtimes accompany their tables; feedback runtimes are omitted because execution conditions differed. All evaluated implementations allocate a single item per request. The general slate formulation specifies the mathematical extension, rather than an additional implemented benchmark.

S2. Reinforcement Background

These classical results motivate exposure constraints and interpret the diagnostic simulations. They are separate from the finite-window serving guarantee. For a fixed catalog of $n \geq 2$ creators, let $f_t(c) > 0$ be reinforced counts, $F_t = \sum_c f_t(c)$, and $\rho_t(c) = f_t(c)/F_t$, where t counts selections. The exponent β controls reinforcement strength; $\sigma_{u,t}$ denotes the corresponding normalized served counts for user u.

Theorem S1 (Classical concentration trichotomy [3])**.** *With unit initial counts and selection proportional to $f_t(c)^\beta$, $\beta \geq 0$: (i) $\rho_t \to (1/n, \ldots, 1/n)$ almost surely for $\beta < 1$; (ii) $\rho_t \to \rho_\infty \sim \mathrm{Dirichlet}(\mathbf{1})$, the uniform law on the probability simplex, for $\beta = 1$, with $\mathbb{E}\,\mathrm{HHI}(\rho_\infty) = 2/(n+1)$; and (iii) for $\beta > 1$ one creator is selected at all but finitely many steps almost surely, so $\mathrm{HHI}(\rho_t) \to 1$.*

Proof outline. For $0 \leq \beta < 1$, the mean ordinary differential equation (ODE) is $\dot{\rho}_c = \rho_c^\beta/S - \rho_c$, $S = \sum_c \rho_c^\beta$. Let D^+ denote the upper right Dini derivative. For $M = \max_c \rho_c$, $m = \min_c \rho_c$, $D^+ \log(M/m) \leq (M^{\beta-1} - m^{\beta-1})/S < 0$ away from uniformity. Boundary faces repel interior trajectories. The sublinear-urn stochastic-approximation result gives almost-sure convergence [3]. At $\beta = 1$, the Pólya limit is Dirichlet and $\mathbb{E}\rho_\infty(c)^2 = 2/[n(n+1)]$. For $\beta > 1$, let $\mathrm{Exp}(1)$ denote the unit-rate exponential law. Rubin's embedding assigns independent waiting times $E_{c,k}/k^\beta$, $E_{c,k} \overset{\text{iid}}{\sim} \mathrm{Exp}(1)$. Each explosion time $T_c = \sum_k E_{c,k}/k^\beta$ is finite almost surely. The times are independent, continuous, and hence distinct. Before the smallest explosion time, every other clock fires only finitely often. This gives (iii). □

Remark S1. The linear limit is random non-uniformity, rather than a few-winner oligopoly. A few-winner configuration in a finite simulation is a transient; the superlinear asymptotic limit is monopoly. The expected factor $2n/(n+1)$ is not a pathwise HHI ratio.

Theorem S2 (Per-user exposure narrowing)**.** *For a user's fixed $K_u \geq 2$ candidates and unit initial counts, the same urn has $H(\sigma_{u,t})$ converging almost surely to a random value below $\log K_u$ at $\beta = 1$, and to zero at $\beta > 1$.*

Proof. Apply Theorem S1 to this user's served counts and use continuity of entropy. The continuous Dirichlet law gives probability zero to exact uniformity. For positive request weights, $\rho = \sum_u q_u \sigma_u$ also implies $\mathrm{HHI}(\rho) \leq \sum_u q_u \,\mathrm{HHI}(\sigma_u)$ by Jensen. If global exposure is supported on S, then each positive-weight user's exposure is supported on S and $H(\sigma_u) \leq \log|S|$. High HHI alone does not restrict support. □

Theorem S3 (Bounded count-only shaping in the superlinear regime)**.** *For $\beta > 1$, a time-homogeneous bonus $b_t(c) = g_c(f_t(c))$ with $|g_c(k)| \leq B_{\mathrm{sh}}$ preserves monopoly under Boltzmann temperature $\tau > 0$, which scales the bonus through $\exp(b_t(c)/\tau)$.*

Proof. The weights are $\widetilde{w}_c(k) = k^\beta e^{g_c(k)/\tau}$ and $\sum_k 1/\widetilde{w}_c(k) \leq e^{B_{\mathrm{sh}}/\tau} \sum_k k^{-\beta} < \infty$. The creator-specific exponential embedding above applies. □

Count-only dependence matters. At $\beta = 1$, the bounded state-dependent two-creator bonus $b_i(\rho) = -\gamma(\rho_i - 1/2)$ with strength $\gamma > 0$ creates mean reversion; boundedness alone is therefore insufficient for this conclusion.

Proposition S1 (Marginal cumulative-entropy bonus)**.** *For bonus weight $\lambda \geq 0$ and $b_t(c) = \lambda\, \partial H(\rho_t)/\partial f_t(c)$, $b_t(c) - b_t(c') = -(\lambda/F_t)\log[f_t(c)/f_t(c')]$.*

Proof. Differentiate $H(f/F) = \log F - F^{-1}\sum_c f_c \log f_c$. The creator-independent terms cancel. Positive unit counts bound the difference by $O(\log F_t/F_t) \to 0$. This derivative of cumulative-count entropy differs from ordinary policy entropy regularization. □

Theorem S4 (Log-share threshold)**.** *For $\beta \geq 0$, $\lambda \geq 0$, $\tau > 0$, and positive initial counts, $b_t(c) = -\lambda \log \rho_t(c)$ gives exponent $\beta_{\mathrm{eff}} = \beta - \lambda/\tau$. Balance holds for $\beta_{\mathrm{eff}} < 1$, a random non-uniform limit for $\beta_{\mathrm{eff}} = 1$, and monopoly for $\beta_{\mathrm{eff}} > 1$. Thus $\lambda^\star = \tau(\beta - 1)$ is the boundary.*

Proof. The factor $F_t^{\lambda/\tau}$ cancels in $f_t(c)^\beta \rho_t(c)^{-\lambda/\tau}$. Apply Theorem S1 for nonnegative effective exponents and Lemma S1 below otherwise. With non-unit initial counts, the linear Dirichlet parameters are those counts; the other regimes are unchanged. □

Lemma S1 (Nonpositive reinforcement)**.** *For finitely many creators, positive initial counts and weights f_i^ν, $\nu \leq 0$, the normalized counts converge almost surely to uniformity.*

Proof. Write $\pi_i = \rho_i^\nu / \sum_k \rho_k^\nu$, $u = (1/n, \ldots, 1/n)$, and $V = \|\rho - u\|_2^2$. Opposite ordering yields $\sum_{i,k}(\rho_i - \rho_k)(\rho_i^\nu - \rho_k^\nu) \leq 0$, hence $\langle \rho, \pi \rangle \leq 1/n$. For the next draw I, let e_I be its unit coordinate vector. Then $\rho' = (F\rho + e_I)/(F+1)$, where primes denote the next state, so

$$\mathbb{E}[V' \mid \rho] \leq (F/(F+1))^2 V + (1 - 1/n)/(F+1)^2.$$

Adding the tail sum of the summable last term makes a nonnegative supermartingale. It converges, and the sum of its expected decreases is finite. Thus V converges and $\sum_t [1 - (F_t/(F_t+1))^2] V_t < \infty$ almost surely. Since the coefficients have divergent sum, the limit of V is zero. □

Algorithm S1 PD: a projected primal–dual solver for CWOP

Require: Request probabilities q_u, admissible slates $\mathcal{A}_u$, exposure incidences $z_{u,a,c}$, welfare $r_{u,a}$, targets (η, κ, h_0), exploration floor ϵ, dual bound B

Ensure: Averaged occupancy $\bar{d}_T$ and conditional serving policy

Define $\mathcal{X}_\epsilon$, $d = \mathcal{Q}x$ and residual vector $g(d) = (\mathrm{HHI}(R(d)) - \eta, (R_c(d) - \kappa)_c, (h_0 - H(S_u(d)))_u)$

Let $\Pi_{\mathcal{S}}$ denote Euclidean projection onto $\mathcal{S}$; γ_t^x, γ_t^ξ are step sizes and T is the iteration count

Initialize $x_1 \in \mathcal{X}_\epsilon$, dual vector $\xi_1 \leftarrow 0$, and running average $\bar{x}_0 \leftarrow 0$

for $t = 1, \ldots, T$ **do**

  Compute the request-slate marginal score $G_t \leftarrow \nabla_x [W(\mathcal{Q}x_t) - \xi_t^\top g(\mathcal{Q}x_t)]$

  $x_{t+1} \leftarrow \Pi_{\mathcal{X}_\epsilon}(x_t + \gamma_t^x G_t)$ ▷ primal ascent

  $\xi_{t+1} \leftarrow \Pi_{[0,B]^{M_g}}(\xi_t + \gamma_t^\xi g(\mathcal{Q}x_t))$ ▷ dual descent

  $\bar{x}_t \leftarrow (1 - 1/t)\bar{x}_{t-1} + x_t/t$

end for

Set $\bar{d}_T \leftarrow \mathcal{Q}\bar{x}_T$ and obtain x^{serve} by repairing $\bar{x}_T$ (Proposition M.1)

for all arriving requests of type u **do**

  Sample an admissible slate $a \sim x_u^{\mathrm{serve}}$

  Log realized position-weighted exposure for tracking

end for

Return $\bar{d}_T$, x^{serve}, and the empirical constraint report

The controller is request-aware: entropy dual effects depend on u and slate contents. Theorem M.1 bounds optimization error; Theorem M.2 certifies finite-window serving.

a) Two-state scope: If popularity f_c and exposure e_c are distinct, cancellation is no longer an identity. For limiting aggregate conversion rate q_{conv}, $\sum_c f_c / \sum_c e_c \to q_{\mathrm{conv}}$ does not establish creator-wise locking. Section S6 specifies the coupled simulation protocols as numerical diagnostics.

S3. Optimization Proofs

A. Exploration, Dual Certificates, and Feasibility Repair

We prove Proposition M.1.

Proof. Mix an unrestricted constrained optimizer with x^0 using weight ϵ/ϵ_0. Every admissible coordinate is at least ϵ, and convexity preserves feasibility. Linearity gives (M.6). Slater's condition and compactness imply strong duality. For any optimal multiplier, $W_\epsilon^* = \max_{x \in \mathcal{X}_\epsilon} \mathcal{L}_x(x, \xi^*) \geq W_0 + \delta_0 \|\xi^*\|_1$, giving (M.7). Finally, convexity gives $g_j(\mathcal{Q}x^{\mathrm{rep}}) \leq (1-\theta)v - \theta\delta_0 = 0$ for each constraint. The welfare identity $W(\mathcal{Q}x^{\mathrm{rep}}) = (1-\theta)W(\mathcal{Q}x) + \theta W_0$, combined with the restricted optimization bound and (M.6), proves the last claim. In computation we add a small positive safety cushion. Bisection along the same segment can reduce the sufficient mixing weight, provided every residual is checked directly; this is not an optimal projection onto the full feasible set. □

B. Convexity of the CWOP Program and Welfare-Gap Accounting

Convexity. The request-occupancy polytope $\mathcal{D}$ is a product of scaled simplices and hence is compact and convex. Every q_u is fixed and positive, making $R(d)$ and $S_u(d)$ affine in d. Therefore $\mathrm{HHI}(R(d)) = \|R(d)\|_2^2$ and $\|R(d)\|_\infty$ are convex, while $H(S_u(d))$ is concave. The HHI and cap sub-level sets and entropy super-level sets are consequently convex. The objective W is linear, so (M.5) is a convex optimization problem. □

Existence and accounting. By Assumption M.1, $\mathcal{F}$ is nonempty. As a closed subset of compact $\mathcal{D}$, it is compact. Continuity of W gives an optimizer by the extreme-value theorem. Since $\mathcal{F} \subseteq \mathcal{D}$,

$$W(\hat{d}) \leq W_g^*, \qquad \Lambda(\mathcal{F}) := W_g^* - W(\hat{d}) \geq 0.$$

This definition gives the claimed identity; it does not supply a numerical welfare bound without specifying and estimating W. □

C. Barrier Limit and Dual Adjustments

a) Barrier characterization: On the strictly feasible exploration-restricted domain, the logarithmic barrier has objective gap at most $M_g\mu$, where $\mu > 0$ is the barrier weight. Its induced dual multipliers remain bounded and every accumulation point is primal–dual optimal. The derivation also identifies how each constraint changes the request-level score.

Proof. Work on $\mathcal{X}_\epsilon$ and remove identically zero exposure coordinates before differentiating entropy. Write $f(x) = W(\mathcal{Q}x)$ and $g_x(x) = g(\mathcal{Q}x)$. A barrier maximizer maximizes $f(x) + \mu \sum_j \log(-g_{x,j}(x))$ over $\mathcal{X}_\epsilon$ with $g_x < 0$. Existence follows from compactness and strict feasibility: the barrier tends to minus infinity at a constraint boundary. Setting $\xi_{\mu,j} = \mu/(-g_{x,j}(x_\mu))$, first-order concavity gives $\sup_{x \in \mathcal{X}_\epsilon} \mathcal{L}_x(x, \xi_\mu) = \mathcal{L}_x(x_\mu, \xi_\mu) = f(x_\mu) + M_g\mu$. Thus $0 \leq W_\epsilon^* - f(x_\mu) \leq M_g\mu$. A strict witness bounds $\delta_0 \|\xi_\mu\|_1 \leq f(x_\mu) + M_g\mu - f(x^0)$, so multipliers remain bounded. Every accumulation point is primal-dual optimal; convergence of the entire pair requires uniqueness, not merely strict complementarity. In x coordinates the HHI and cap gradient terms are $2q_u \sum_c R_c z_{u,a,c}$ and $q_u z_{u,a,c}$, whereas the entropy gradient is $\sum_c z_{u,a,c}(-\log S_{u,c} - 1)$. Consequently entropy adjustments depend on the request and slate; they are not generally a single creator-only price. The barrier limit concerns the restricted problem; the unrestricted comparator is covered by Proposition M.1. □

D. Proof of Theorem M.1

Proof. The saddle function $\mathcal{L}_x(x, \xi) = W(\mathcal{Q}x) - \xi^\top g(\mathcal{Q}x)$ is concave in x and affine in ξ. The standard projected-gradient inequality [4–6], summed for primal ascent and dual descent with steps proportional to $t^{-1/2}$, gives for every $(x, \xi) \in \mathcal{X}_\epsilon \times \Xi_B$

$$\frac{1}{T} \sum_{t=1}^{T} \big(\mathcal{L}_x(x, \xi_t) - \mathcal{L}_x(x_t, \xi) \big) \leq \varepsilon_T,$$

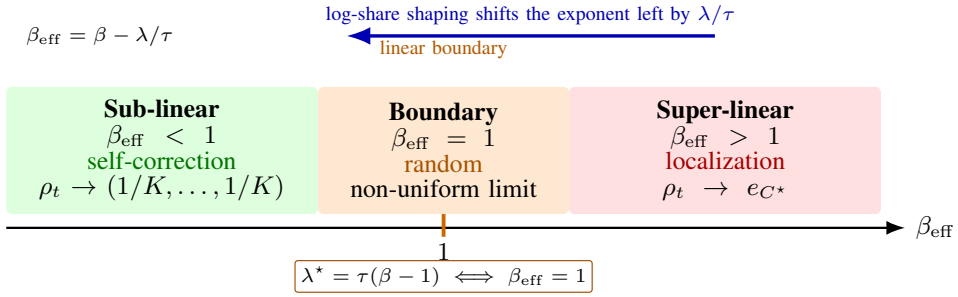


Fig. S1. Three feedback regimes and the log-share threshold (schematic). Within the single-state power-law model, log-share shaping shifts the exponent exactly by λ/τ. The limit at the linear boundary is non-uniform but need not be an oligopoly.

with ε_T as stated. Concavity, convexity, and Jensen's inequality then yield

$$\max_{x \in \mathcal{X}_\epsilon} \mathcal{L}_x(x, \bar{\xi}_T) - \min_{\xi \in \Xi_B} \mathcal{L}_x(\bar{x}_T, \xi) \le \varepsilon_T.$$

Let $d_\epsilon^\star$ be a constrained optimum. Since it is feasible, $\mathcal{L}_x(\mathcal{Q}^{-1} d_\epsilon^\star, \bar{\xi}_T) \ge W_\epsilon^*$. Selecting $\xi = 0$ in the gap bound gives $W_\epsilon^* - W(\bar{d}_T) \le \varepsilon_T$.

For constraint violation, let $v_j = \mathbf{1}\{g_j(\bar{d}_T) > 0\}$ and select $\xi = \xi^* + r_\xi v$, which lies in Ξ_B by assumption. Strong duality gives $W_\epsilon^* - W(\bar{d}_T) + (\xi^*)^\top g(\bar{d}_T) \ge 0$. Substitution in the saddle-gap inequality leaves $r_\xi \sum_j [g_j(\bar{d}_T)]_+ \le \varepsilon_T$, proving the second bound.

For serving, let N_u count requests of type u. The exposure credited to creator c on each independent request is a random variable in $[0, 1]$ with mean $R_c(\bar{d}_T)$. Hoeffding's inequality and a union bound over creators give the stated aggregate coordinate bound. Conditional on N_u, applying the same argument to every creator coordinate for type u and taking a union bound over types gives the ℓ_1 bound for $\widehat{S}_u$. The maximum-share error is at most the maximum coordinate error, and

$$\begin{aligned} |\operatorname{HHI}(\widehat{R}) - \operatorname{HHI}(R)| &= \left| \sum_c (\widehat{R}_c - R_c)(\widehat{R}_c + R_c) \right| \\ &\le 2 \|\widehat{R} - R\|_\infty. \end{aligned}$$

Write $h_2(v) = -v \log v - (1-v) \log(1-v)$ for binary entropy. Finally, with $\Delta_u = \frac{1}{2} \|\widehat{S}_u - S_u\|_1$, the Fannes-Audenaert bound $|H(\widehat{S}_u) - H(S_u)| \le \Delta_u \log(|\mathcal{C}_u| - 1) + h_2(\Delta_u)$ (or the trivial $\log |\mathcal{C}_u|$ bound outside its usual range) proves entropy tracking as $N_u \to \infty$. □

E. Proof of Lemma M.1

Proof. Base feasibility implies $0 \le R_c \le \kappa$, hence $R_c(1 - R_c) \le \kappa$ and $A_R \le \kappa$. For $Y = R_I \in [0, \kappa]$, $Y^2 \le \kappa Y$ gives $\operatorname{Var}(Y) \le \mathbb{E}Y(\kappa - \mathbb{E}Y) \le \kappa^2/4$. Substitution in (M.11), together with $m_u \le \min\{\log K_u, b_u\}$, proves the componentwise envelope. Mix a base optimum with x^0 using $\bar{\vartheta}$. Convexity gives $g_i(\mathcal{Q}x) \le -\bar{\vartheta} s_i \le -\bar{m}_i$. The mixture is base-feasible, so its own margins obey $m_i(x) \le \bar{m}_i$ and it passes the certificate. Its score is the claimed lower bound. The same argument applies to any base-feasible policy of score W_{lb}. No lower bound from this witness is asserted when $\bar{\vartheta} > 1$. □

F. Independent-Serving Tracking

The following bounds accompany Theorem M.1. If N independent request-slate pairs are sampled from the conditional policy represented by $\bar{d}_T$, let $\widehat{R}$ and $\widehat{S}_u$ be the empirical aggregate and per-type exposure shares. For a failure budget $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\max_c |\widehat{R}_c - R_c(\bar{d}_T)| \le \sqrt{\frac{\log(2|\mathcal{C}|/\delta)}{2N}}. \tag{S1}$$

Let $C_\Sigma = \sum_u |\mathcal{C}_u|$. Separately, conditional on the type counts $N_u > 0$ (the number of requests of type u), with probability at least $1 - \delta$ a simultaneous coordinate bound gives $\|\widehat{S}_u - S_u(\bar{d}_T)\|_1 \le |\mathcal{C}_u| \sqrt{\log(2C_\Sigma/\delta)/(2N_u)}$. Consequently maximum-share and HHI errors vanish at $O(N^{-1/2})$. The Fannes-Audenaert bound implies that every entropy error vanishes as $N_u \to \infty$ [7].

S4. Adaptive Serving Proof

A. Proof of Theorem M.2

Proof. Condition on the history before a window, fixing its policy. Single-item exposures are independent categorical vectors with mean R; write $e = \widehat{R} - R$. Allocate $\delta/6$ to each of six event families over all windows: caps, the two HHI terms, type counts, and two entropy terms.

Maximum share. Each creator indicator has variance $R_c(1 - R_c)$ and range one. One-sided Bernstein gives $e_c \le m_c$ over all JC coordinates, outside probability $\delta/6$.

HHI linear term. Expand

$$\operatorname{HHI}(\widehat{R}) - \operatorname{HHI}(R) = 2\langle R, e \rangle + \|e\|_2^2.$$

Writing C_i for the creator served on request i, each summand R_{C_i} has variance V_R and range A_R. One-sided Bernstein bounds $2\langle R, e \rangle \le 2\sqrt{2V_R \ell_H / N} + 4A_R \ell_H/(3N)$ outside total probability $\delta/6$ over J windows.

HHI quadratic term. Jensen gives $\mathbb{E}\|e\|_2 \le \sqrt{(1 - \operatorname{HHI}(R))/N} \le 1/\sqrt{N}$. Replacing one observation changes $\|e\|_2$ by at most $\sqrt{2}/N$. Bounded differences imply $\|e\|_2 \le (1 + \sqrt{\ell_H})/\sqrt{N}$ outside another total probability $\delta/6$. Combining the HHI events gives m_H.

Type counts. Let $N_{j,u}$ count type-u requests in window j. The binomial Chernoff bound gives $N_{j,u} \geq n_u$ except with probability at most $e^{-\ell_U}$ for each pair, hence at most $\delta/6$ overall.

Per-type entropy, bounded-difference branch. Condition also on the type sequence. For n categorical observations from p on at most K points, the plug-in entropy bias satisfies

$$0 \leq H(p) - \mathbb{E}H(\widehat{p}) = \mathbb{E}D(\widehat{p}\|p) \leq \log(1 + (K-1)/n).$$

Indeed, $D(v\|p) \leq \log \sum_i v_i^2/p_i$ and Jensen apply, with $\mathbb{E}\sum_i \widehat{p}_i^2/p_i = 1 + (|\operatorname{supp} p| - 1)/n$; omit zero-probability coordinates. This is the usual plug-in bias bound [8]. Moving one observation between bins changes entropy by at most $[n\log n - (n-1)\log(n-1)]/n \leq (\log n + 1)/n$. One-sided bounded differences therefore give

$$H(p) - H(\widehat{p}) \leq \log(1 + (K-1)/n) + (\log n + 1)\sqrt{\ell_U/(2n)}.$$

Both terms decrease for $n \geq 3$, so substitute n_u on the sufficient-count event. The trivial bound $\log K$ permits clipping. This branch costs at most $\delta/6$ over all JU pairs.

Per-type entropy, policy-dependent branch. For $p_i > 0$, direct expansion gives

$$H(p) - H(\widehat{p}) = \sum_i (\widehat{p}_i - p_i)\log p_i + D(\widehat{p}\|p).$$

The linear term is a mean of centered $\log p_I$, with variance $V(p)$ and range $A(p)$. Bernstein bounds it by $\sqrt{2V(p)\ell_U/n} + 2A(p)\ell_U/(3n)$. For any empirical type v, the multinomial theorem gives $\Pr(\widehat{p} = v) \leq \exp[-nD(v\|p)]$. There are at most $(n+1)^{K-1}$ types: the first $K-1$ counts each lie in $\{0,\dots,n\}$ and determine the last. Consequently

$$\Pr\left(D(\widehat{p}\|p) > \frac{(K-1)\log(n+1) + \ell_U}{n}\right) \leq e^{-\ell_U}.$$

Both linear and KL bounds decrease with $n \geq 3$, so replace n by n_u. The two event families cost $2\delta/6$ overall. For each type and history, select the smaller of the bounded-difference and policy-dependent branches before sampling. The selected branch has failure probability at most $2e^{-\ell_U}$, without paying for both branches. Clipping at $\log K$ is deterministic.

These bounds are uniform in the conditioned history; the tower property and union bound therefore apply under adaptive replanning. The six failure allocations sum to δ. The margins absorb the one-sided deviations. Global pooled exposure preserves the HHI and cap bounds by convexity. For each type, its pooled exposure uses its realized window-count weights; entropy concavity preserves its floor. □

B. Model Error and Simulation-Based Calibration

The request law and sampling model must match the plan. Arbitrary history-dependent scores include the experiment's reinforcing scores, but the theorem does not establish their convergence. Define total variation as $\mathrm{TV}(p,p') = \frac{1}{2}\|p - p'\|_1$ and let R_q denote aggregate exposure under request law q. If a separately validated estimate $\widehat{q}$ satisfies $\mathrm{TV}(q,\widehat{q}) \leq e_q$, a fixed policy has $\|R_q - R_{\widehat{q}}\|_\infty \leq e_q$. Additional reserves e_q for the cap and $2e_q$ for HHI handle this mismatch; count bounds need valid lower bounds on true q_u. Confidence failures for those estimates must also be budgeted. Unknown-law learning is not evaluated here.

Finite Monte Carlo estimates are not exact safety guarantees. A certificate based on sampling a known policy needs independent validation, one-sided binomial confidence bounds, and allocation of validation and serving failures across windows and searched policies. Ordinary bootstrap margins do not automatically meet these requirements. The certified experiment uses analytic margins without outcome-based selection.

C. General Slates and a Total-Variation Alternative

a) Policy-independent single-item certificate: When a margin is needed before a policy is available, use $\ell'_C = \log(5JC/\delta)$, $\ell'_H = \log(5J/\delta)$ and $\ell'_U = \log(5JU/\delta)$, with the corresponding count lower bounds $n'_u \geq 3$. Primes mean that ℓ'_U replaces ℓ_U in the definitions of n_u and b_u. For plans with $R_c \leq \kappa$, sufficient margins are

$$m'_C = \sqrt{2\kappa\ell'_C/N} + 2\ell'_C/(3N),$$
$$m'_H = \kappa\sqrt{2\ell'_H/N} + (1 + \sqrt{\ell'_H})^2/N,$$

and the bounded-difference entropy margin $\min\{\log K_u, b'_u\}$. The proof uses $R_c(1-R_c) \leq \kappa$, Hoeffding for $R_{C_i} \in [0,\kappa]$, and five event families. This is a sufficient feasibility threshold; the utility and witness-mixing curves in Figure M.4 describe its cost. Separate per-type entropy floors avoid replacing every floor by the largest entropy margin.

b) General slates: Normalized position-weighted slates contribute simplex-valued vectors rather than categorical vertices. A general certificate uses $\epsilon_R = \sqrt{\log(6JC/\delta)/(2N)}$ and

$$v_u = \min\left\{1, \frac{K_u}{2}\sqrt{\frac{\log(6J\sum_v K_v/\delta)}{Nq_u}}\right\}.$$

For $Nq_u \geq 8\log(3JU/\delta)$, coordinate Hoeffding bounds, the event $N_{j,u} \geq Nq_u/2$, and three union bounds give margins $(2\epsilon_R, \epsilon_R, \omega_{K_u}(v_u))$. Here $\omega_K(v) = v\log(K-1) + h_2(v)$ for $v \leq 1 - 1/K$, and $\log K$ otherwise, where h_2 is binary entropy [7]. The HHI step uses $|\operatorname{HHI}(p) - \operatorname{HHI}(p')| \leq 2\|p - p'\|_\infty$. It applies to a broader class of exposure vectors than the categorical bounds above.

Alternatively, total variation $\mathrm{TV}(p,p') = \frac{1}{2}\|p - p'\|_1$ is the maximum discrepancy over nonempty proper subsets. One-sided Hoeffding for their $2^{K_u} - 2$ subset masses yields

$$\Pr(\mathrm{TV}(\widehat{S}_u, S_u) > v \mid N_u = n) \leq (2^{K_u} - 2)e^{-2nv^2}.$$

This gives square-root rather than linear dependence on K_u, as in the classical finite-alphabet inequality [9]. The subset proof also covers simplex-valued slate contributions. The single-item theorem instead bounds categorical entropy directly, avoiding the additional continuity conversion.

D. Proof of Local Cap Necessity

For Proposition M.2:

Proof. The creator count is binomial (N, p_N). Its centered, variance-normalized sum converges to a standard normal because $p_N \to \kappa \in (0,1)$. The cap threshold converges to z on this scale; lattice rounding vanishes. The final claim follows by applying the same argument to any subsequence with a smaller threshold. □

E. Proof of Corollary M.1

Proof. Mix an untightened optimizer with x^0 in proportion ϑ. Convexity gives residual at most $-\vartheta s_i \le -\Delta_i$, preserving the exploration floor. Linearity gives loss $\vartheta(W_\epsilon^* - W_0)$. The tightened optimum can only improve this construction. The certificate does not apply if a required margin exceeds the witness slack. □

Together with Proposition M.1, this separates target cost, sampling safety, exploration, and numerical error for the same window's score. Different controllers induce different future scores; their end-to-end utility difference is not this fixed-window budget or a regret bound.

a) Interpreting the utility budget: Corollary M.1 bounds loss for one fixed window's score by mixing with a strict witness. Its proof is in Section S4.E. This gives a fixed-history score guarantee. Comparisons across feedback trajectories add the effects of controller-induced changes in future scores.

b) Interpreting the certified-score bound: The witness-mixture proof is in Section S3.E. The lemma supplies an attainable lower bound on the best certified score. Refinement builds on this guarantee without requiring global optimization of the certified set. An exploration floor alone does not determine which entropy branch is smaller: witness mixing can move all probabilities far above the floor.

S5. Experimental Protocols

These experiments trace the joint-control argument from numerical feasibility to realized exposure: optimization checks support Q1, feedback and sampling tests support Q2, listening-derived candidates support Q3, and public short-video creator candidates support Q4. The ablations identify which parts of CWOP produce joint feasibility, utility retention, and serving safety.

A. Independent Optimum, Feasibility, and Error Accounting

The computational implementation separates the original controller, a vectorized eligible-edge implementation, and an independently formulated CVXPY conic model. The conic variables are the eligible single-item probabilities; row sums, exposure sums, quadratic HHI, and exponential-cone entropy constraints are assembled separately. Clarabel uses absolute/relative gap and feasibility tolerances 10^{-10} and a 300-iteration cap. The Splitting Conic Solver (SCS) uses tolerance 2×10^{-7} and at most 200,000 iterations on the base instance. The reference bounds use Clarabel.

For nonnegative returned constraint multipliers ξ and a strictly positive feasible point y, concavity of $\mathcal{L}_x$ gives the independently computed upper bound

$$\max_{x \in \mathcal{X}_\epsilon,\, g(\mathcal{Q}x) \le 0} W(\mathcal{Q}x) \le \mathcal{L}_x(y,\xi) + \max_{x \in \mathcal{X}_\epsilon} \langle \nabla_x \mathcal{L}_x(y,\xi), x - y \rangle.$$

The last maximum is rowwise linear optimization on a floored simplex. A repaired feasible policy supplies the lower bound. Computation uses a 10^{-9} mixture with uniformity to keep the tangent point strictly positive. These are floating-point numerical certificates, not interval-arithmetic proofs. Solver statuses and checked upper–lower gaps are recorded during a run.

The controller's raw average can be infeasible. We first compute the sufficient mixture weight from Proposition M.1, then bisect along the segment toward a strict uniform-over-candidates witness, requiring every residual to be at most -10^{-10}. This reduces needless mixing without claiming an optimal projection. The oracle and every reported feasible baseline undergo the same residual checks. Let $\mathcal{N}(\mu, \sigma^2)$ denote the normal law. The original synthetic utility generator uses creator effects $\mathcal{N}(0, 0.12^2)$, shared-head bonuses linearly spaced from 1.35 to 0.90, group bonuses 0.85, and independent user–creator noise $\mathcal{N}(0, 0.10^2)$. Request weights are independent Uniform$(0.65, 1.35)$ draws, normalized to sum to one. The base graph gives every type six shared head creators, five group creators, and seven sampled alternatives. The larger graph has 48 types, 72 creators, 30 candidates, eight head creators and eight groups. Its targets are $(2/72, 2.88/72, 0.65 \log 30)$. The runners save sampled utilities, masks, request probabilities, and computed policies locally.

Base graph seeds are 20260722–20260726; the larger graph uses 20260752. PD uses $\epsilon = 10^{-7}$, primal rate $0.75/\sqrt{t}$, and dual rate $1.25/\sqrt{t}$; the global-only ablation uses $5/\sqrt{t}$. The feasible softmax envelope searches temperatures $\{0.01, 0.05, 0.1, 0.2, 0.3, 0.5, 1, 2\}$ and applies exact segment feasibility checks. It optimistically chooses the best feasible utility on each instance, rather than claiming a held-out tuning result. Global-only and local-only are diagnostic ablations, not competitive joint-constraint methods. The frontier protocol solves the constrained problem independently at entropy floors $\{0.30, 0.45, 0.55, 0.65, 0.75, 0.85\}$; it does not connect infeasible controller iterates.

B. Repeated Feedback and Comparator Definitions

The response model and frozen-window feedback score are specified in Section M.VII-D. We give the split, tuning and comparator details below. Impressions update the exposure counts used by the log-share and FairCo adaptations; sampled clicks update popularity and affinity.

Synthetic training graphs use seeds 20260901 and 20260902; tests use 20260910–20260914. Last.fm tuning uses SVD splits 20260916 and 20260917, disjoint from test splits 20260920–20260922. Every method receives 20 windows of 20,000 requests on synthetic graphs and 100,000 on Last.fm

graphs. A deterministic seed by graph and window governs request counts, creator counts and responses. Different policies induce different outcomes. Tuning selects highest mean intrinsic utility among settings with at most 5% individual-window violations across both training trajectories. If none qualify, minimize violation fraction, then mean normalized excess, then maximize utility. The normalized excess in a window is $\max\{0, \mathrm{HHI}(\widehat{R})/\eta - 1, \max_c \widehat{R}_c/\kappa - 1, (h_0 - \min_u H(\widehat{S}_u))/h_0\}$. A violation means a raw residual exceeding 10^{-6}, with entropy in nats. The original four-baseline comparison has 46 training trajectories per dataset. Four selected baselines and two conic policies give 30 synthetic and 18 Last.fm test trajectories. All outcomes and final policies are generated by the runners. The comparator definitions are:

- *Temperature*: policy entropy regularization, with temperature searched over $\{0.15, 0.3, 0.6, 1.2, 2.4, 4.8, 9.6, 19.2\}$.
- *Log-share*: subtract λ times both the global and local log-exposure ratios, using one pseudocount per eligible creator, then softmax at temperature 0.1. Search $\lambda \in \{0.05, 0.15, 0.3, 0.6, 1.2\}$.
- *FairCo-style batch*: singleton creator groups with known merit $M_c = \max\{10^{-6}, \sum_u q_u p_{u,c} \mathbf{1}\{c \in \mathcal{C}_u\}\}$. Add $\lambda[\max_k E_k/M_k - E_c/M_c]$ to the current score, with cumulative exposures E, then choose the top item (floor 10^{-7}). Search $\{10^{-6}, 10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}\}$. This adapts the exposure-disparity score in FairCo [10], but freezes updates for an entire window, uses a biased score and known merits, and does not reproduce its online estimator. Its native merit-proportionality target differs from our HHI/entropy intersection, so this comparison evaluates the adaptation on our specified targets.
- *FairCo + certificate*: apply the checked witness-segment search to the FairCo-style batch proposal before every serving window. Use the same training-selected correction weight, window size, response model, and random-seed scheme as the unmodified batch comparator. Update feedback using the combined policy's own sampled outcomes. Check the final policy-dependent certificate before sampling; no conic refinement or extra parameter tuning is used. This construction adds the main paper's exposure certificate but does not assert preservation of FairCo's native merit-proportionality guarantee.
- *Two-sided batch FW*: 10,000 conditional-gradient steps, initialized at uniformity, maximize $W_s(x) - AC\,\mathrm{HHI}(R) + A\sum_u q_u H_{10^{-3}}(x_u)$, where $W_s(x) = \sum_{u,c} q_u x_{u,c} s^j_{u,c}$ is the current score objective, A is the diversity-penalty weight, and $H_\zeta(x) = -\sum_c (x_c + \zeta)\log(x_c + \zeta)$ on eligible edges. The vertex selects the largest adjusted score per type, with step $2/(t+2)$. Search $A \in \{0.1, 0.3, 1, 3, 10\}$. This uses the Frank–Wolfe ranking principle of Offr [11], but is a batch, known-q, new-objective adaptation rather than the original online algorithm.
- *Conic reference*: independently solve the same window score objective and original targets using the conic oracle, repair numerical residuals, and deploy it. It supplies the most accurate constrained solutions on these graphs.
- *Conic reference + certificate*: checked reoptimization from Section M.VI, with $J = 20$ and $\delta = 0.05$.

The per-window objective is the reinforced score, whereas reported utility is evaluated using p. All controllers have the same information but different computational budgets; equal iteration counts do not imply equal computational cost. Feedback runtimes are omitted; static optimizer timings are reported separately. FORM provides related unknown-reward learning guarantees beyond this known-prior protocol.

a) Paired utility comparison protocol: We compare the selected methods on the same held-out graph or listening split, request law, exposure targets, and window size. Each method follows its own feedback trajectory. Let U_i^{CWOP} and U_i^{FW} denote the final pooled intrinsic utilities on instance i, measured as in Table M.III. The mean gain is $100(\bar{U}_{\mathrm{CWOP}}/\bar{U}_{\mathrm{FW}} - 1)$, where bars average over the five synthetic graphs or three Last.fm splits. Per-instance gains use $100(U_i^{\mathrm{CWOP}}/U_i^{\mathrm{FW}} - 1)$.

The unit of comparison is a graph or split; windows within a trajectory are not counted as independent replicates. The Last.fm comparisons use repeated splits of the same public dataset. The code compares every selected baseline after the matched experiments have been rerun. Result files are not bundled.

The same ratio-of-means and per-instance calculations apply to the additional joint PG comparator and its certificate wrapper. For KuaiRand, the three test user groups are disjoint but share creator selection and proxy fitting; their SD is descriptive across groups rather than evidence from independent deployments. The strongest zero-violation baseline is identified separately in each dataset.

b) Constraint-specific outcome accounting: Constraint-specific accounting uses the same matched-window trajectories. Global failure means empirical HHI or maximum share exceeds its own target by more than 10^{-6}; local failure means any type's entropy falls below its floor by more than 10^{-6} nats. Concentration columns maximize each window's ratio to its graph-specific target, and the entropy column minimizes $H/\log K$ over all types and windows. These worst-window quantities differ from pooled averages. The replay script regenerates request, creator, and click draws using the original sampling seeds, then checks the window metrics and constraint-specific counts.

c) Additional 10k-window diagnostics: Separate synthetic ablations use $20 \times 10{,}000$ requests to compare PD and the conic reference and vary feedback strength. PD uses 10,000 updates per window. An uncontrolled comparison uses softmax temperature 0.1. Empirical margins $(0.005, 0.01, 0.07\log 18)$ are fixed before serving. These experiments isolate solver error and margin sensitivity; the matched-window certified comparisons above all use Theorem M.2. Their role is to distinguish protection of exposure targets from the choice of solver and from the strength of the feedback score. Feasibility without margins is a property of the planned distribution. A window is counted as violating when its largest raw residual exceeds 10^{-6} (entropy residuals are

in nats). The oracle can therefore have optimal planned utility and still violate empirical targets. The pooled distribution uses actual request counts, not the planned q. Clicked-item entropy is reported in Table M.VIII; it is not constrained by CWOP. Reported spreads are sample standard deviations across five graphs. Paired utility intervals use Student's t_4 multiplier on within-seed differences; they quantify this small synthetic design.

d) Public-data optimization tuning: The public artist-listening candidate construction is specified in Section M.VII-E. On the disjoint split 20260919, test primal coefficients $\{0.25, 0.75, 2.25, 6.75\}$ and dual coefficients $\{5, 20, 80, 240\}$ for 5,000 iterations, choosing maximum repaired feasible utility. The selected coefficients $(2.25, 240)$ are then frozen for all test splits, each run for 30,000 iterations. Uniform witnesses certify $B = 200$.

C. Matched-History Target and Certification Costs

`run_fairco_tradeoff.py` takes locally generated matched results as input. The frozen-history diagnostic selects windows 1, 10, and 20 from each of the three certified Last.fm trajectories before inspecting diagnostic outcomes. It replays all preceding request, exposure, and response draws, checking stored scores and window metrics. On each common history, it solves the untightened constrained program and compares its score with the greedy score and a certified plan. The original setting uses its saved certified plan; changed targets use the same checked conic refinement procedure. Thus the decomposition holds for one fixed score function and does not compare different feedback histories.

The prespecified settings are $(a, b) \in \{(1.5, 0.65), (1.5, 0.50), (2.0, 0.65), (2.0, 0.50), (2.5, 0.35)\}$, with HHI and share ceilings equal to a times their uniform-witness values and entropy floor $b \log 64$. Every setting is reported. Each certified policy receives 100 independently sampled validation windows using seeds determined by split, window index, and setting. Validation outcomes never select a plan. Means first average the selected histories within each split, then average the three splits; any reported standard deviation is across splits. The diagnostic reports current-score cost fractions and intrinsic-response utility relative to its unconstrained optimum separately. Frozen-history comparisons isolate target and certification costs; evaluating subsequent feedback effects requires separate trajectories. No numerical results or saved inputs are included in this supplement.

D. Public Short-Video Creator Graph

`prepare_kuairand.py` uses the public KuaiRand-Pure release [2], obtained separately from the authors' linked Zenodo archive. The source archive has published MD5 `0820331067a3784d9691136f772b35a7`; the download instructions require checking it before extraction. The public video-to-author mapping defines creators. Invalid author identifiers are excluded.

Only standard-recommendation interactions from April 8–21, 2022 fit the response proxy. Select the 256 creators with the most distinct users in that period, breaking ties by public identifier. For each user–creator pair, let $n_{u,c}$ and $y_{u,c}$ be impression and positive-response counts. The dataset's binary response denotes a click in its two-column interface and a valid play in its single-column interface. Let $r = \sum y / \sum n$. Smooth creator and user rates with 20 observations at rate r, giving r_c and r_u. Set $B_{u,c} = \operatorname{logit}(r_u) + \operatorname{logit}(r_c) - \operatorname{logit}(r)$ and $\widetilde{p}_{u,c} = (y_{u,c} + 5\,\operatorname{sigmoid}(B_{u,c}))/(n_{u,c} + 5)$. Take a rank-16 SVD approximation of $\operatorname{logit}(\widetilde{p}) - B$ and apply sigmoid after adding B back. Probabilities are clipped to $[10^{-8}, 1 - 10^{-8}]$ only when evaluating logit. The proxy averages over a creator's observed videos, providing a creator-level response model for the controlled simulation.

Eligible users have observed at least five selected creators. Seed 20261112 selects 640 users without replacement and partitions them into five disjoint groups of 128. The first two groups tune controllers; the remaining three are held out. Each user receives its 32 highest-proxy creators and 32 seeded alternatives from the remaining catalog. Group seeds are 20261110, 20261111, and 20261120–20261122. Request probabilities are uniform. Targets are 1.5 times uniform-witness HHI and maximum share and $0.65 \log 64$ entropy. All methods receive $J = 20$ windows of $N = 100{,}000$ requests with $\delta = 0.05$, the same feedback equation and Bernoulli proxy-response model as the matched Last.fm experiment. Baseline grids and training-only selection are unchanged. FairCo and policy-gradient certificate wrappers inherit their unmodified comparator's selected parameter.

Later randomized interactions from April 22–May 8 are used only to check the fixed proxy's AUC and Brier score against a creator-rate predictor. They do not fit the model, choose users or candidates, tune controllers, select targets, or select reported trajectories. The test measures prediction on these logged exposures; the controller experiment still uses simulated responses. Pure retains a restricted video pool and incomplete sequences. Its fixed candidate graph isolates exposure control; changing supply, video recency, and creator entry call for a sequential extension with changing candidates. No inputs, fitted arrays, numerical results, or public identifiers are bundled in the supplement.

E. Policy-Gradient Comparator under Joint Targets

We adapt the stochastic-ranking policy-gradient framework of Singh and Joachims [12] to the same known-score single-item problem. A tabular logit $\theta_{u,c}$ defines a categorical policy $x_{u,c}$ by softmax, the length-one Plackett–Luce model. This parameterization represents every strictly positive policy on the fixed candidate graph, providing a direct optimization comparison for the observed request types. Replace merit

disparity with the three exposure-target families:

$$D(x) = \tfrac{1}{2}\Bigg[[\mathrm{HHI}(R)/\eta - 1]_+^2 + \frac{1}{C}\sum_c [R_c/\kappa - 1]_+^2 + \frac{1}{U}\sum_u [1 - H(x_u)/h_0]_+^2\Bigg].$$

Maximize $W_s(x) - \lambda D(x) + 10^{-3}\sum_u q_u H(x_u)$ using 2,000 Adam steps per window, learning rate 0.03, moment coefficients $0.9, 0.999$, and denominator offset 10^{-8}. Initial logits are zero; later windows warm-start from the selected preceding logits. Select the iterate with the largest penalized training objective, then apply the shared exploration floor 10^{-7}. Search $\lambda \in \{0.1, 1, 10, 100, 1000\}$ on the original two training instances of each dataset using the same violation/utility rule. All five synthetic, three Last.fm, and three public-feed test graphs are retained.

For length-one actions, finite-action expectations and their policy gradients can be evaluated exactly. We use these instead of Monte Carlo gradient estimates. This changes the estimator, parameterization, and fairness penalty relative to the original learning-to-rank implementation, and is labeled *joint PG (Singh–Joachims-style)*. These adaptations align the stochastic-ranking framework with the shared single-item joint-target protocol. The certificate wrapper adds finite-window sampling protection to the raw penalty. *Joint PG + certificate* applies the same checked witness-segment repair, with no conic refinement, and follows its own sampled feedback trajectory. Its penalty inherits the raw comparator's selection. The gradient is independently checked by finite differences; saved policies permit replaying every reported window after local reruns.

F. Confidential Platform Audit Protocol

The main paper reports dimensionless observational summaries from a commercial content platform. Signal names are generic analytical labels. The complete serving extract and the audience-matched creator set are separate populations: signal HHIs use the complete extract, while audience comparisons and cap accounting use the matched set. Each HHI is the sum of squared creator shares for the stated signal.

For audience comparisons, intersect exposed creators with creators present in both snapshots. Define stock as the later audience size and growth as the nonnegative snapshot difference. Resample creators with replacement under a fixed bootstrap protocol, recomputing numerator and denominator on the same sampled creators. Round ratios to one decimal place and interval endpoints outward. Lorenz curves order creators by increasing exposure and compare cumulative shares. The matched set has limited serving coverage, so its ratios characterize linked creators.

The separate audience panel uses the subsequently recorded growth measure, retains zero growth, and pools sparse categories. A Poisson generalized linear model (GLM) includes interval and category effects, with creator-clustered standard errors. The log-stock coefficient is the stock–growth elasticity. The recorded panel outcome differs from the reconstructed snapshot difference, and its window can differ from snapshot spacing. Creator quality, eligibility, and prior recommendations can affect this association; it is descriptive evidence.

For cap accounting, proportional water filling truncates observed creator shares at a cap and rescales the remaining shares to sum to one. Compare HHI, its reciprocal effective creator count, and the share of a fixed leading-creator group with their observed values on the same matched population. Preserve that group across cap settings. This deterministic accounting studies aggregate exposure; candidate eligibility, relevance, and user response enter the controlled CWOP model separately. The relative accounting results appear only in the main paper.

Raw records, original field names, absolute scales, collection dates, exact cap values, and source-specific metadata are excluded. The controlled records retain parsing checks, snapshot alignment, resampling settings, and provenance. Authorized access is needed to reproduce the private audit; the released computational code reproduces the separately specified synthetic and public-data evaluations.

G. Policy-Dependent Safety and Frozen-Policy Tests

These checks support the serving component of the thesis: a joint target is useful only when its certificate is computationally attainable and its utility cost is measured at the intended window size.

a) Larger optimization instance: A separately seeded stress test has 512 request types, $1{,}024$ creators, and 64 candidates per type ($32{,}768$ variables). It uses eight shared-head creators, 32 groups, and the synthetic utility generator above. Targets are 1.5 times the uniform-over-candidates witness's HHI and maximum share, with entropy floor $0.65 \log 64$. The projected solver uses $10{,}000$ iterations and a witness-certified dual box $B = 800$. The protocol separates optimization error from the sampling cost of protecting more request types.

b) Constructing and checking policies: The implementation evaluates every term of Theorem M.2 from the known request probabilities and the proposed policy. The witness is uniform over each request type's candidates. A 64-interval search along the segment from a conic solution to the witness locates a certified point, then 30 bisection steps refine that interval. Every final policy is checked directly (floating-point tolerance 10^{-10}). This search does not assume global monotonicity of the policy-dependent certificate. If no point passes, the procedure reports no certificate. No serving outcomes enter this search.

For both static curves and feedback experiments, up to three additional conic solves use the preceding certified policy's margins and a separate entropy floor for every type. Each candidate is checked again and, when necessary, mixed with the witness. The highest-utility certified construction is retained. The result is a feasible lower bound on attainable certified utility. The fixed common-cap implementation uses $\max_c m_c$ during these extra conic solves, while final validation checks each creator separately. Whenever the uniform envelope fits

the witness slack, the implementation also considers the mixture attaining the computable lower bound of Lemma M.1. The segment-only construction is an ablation. If an optional refinement solve fails, the procedure retains its previously certified plan. Every proposed serving policy is checked before deployment. An exploration lower bound alone does not force the actual probabilities to lie near that bound. Missing curve points indicate that the chosen witness and construction did not yield a certificate. Missing values are not interpolated and do not establish that every policy is infeasible at that size.

c) Variation across synthetic graphs: To evaluate the stability of the sample-size results, we repeat the construction on the five static synthetic graphs with seeds 20260722–20260726. The candidate sets, request probabilities, and scores are generated by the static benchmark. We keep the targets, solver settings, $J = 20$, and $\delta = 0.05$ fixed, and evaluate the same seven request counts from 5,082 to 200,000 on every graph. For each graph, we normalize score gain by its own uniform-witness and conic-reference scores before computing the mean. Figure M.4 reports these means with one sample standard deviation across graphs, using denominator $5 - 1$. The bars measure variation across graph instances; they are not confidence intervals for serving failures. A score point is included only when the construction succeeds on all five graphs. The computational artifact includes the aggregation script; per-graph policies and numerical results are generated locally and are not bundled.

d) Feedback and response generation: The synthetic safety runs use seeds 20260910–20260914, $J = 20$, and $N = 20{,}000$. Last.fm uses test splits 20260920–20260922, $J = 20$, and $N = 100{,}000$. Response probabilities on Last.fm equal the learned proxies clipped to $[0, 1]$. The underlying observations are listening counts; the Bernoulli responses are generated for this experiment. Per-window counts are drawn by first sampling a multinomial type count and then a multinomial creator count within each type. Bernoulli responses are generated by binomial thinning of those counts. This is distributionally equivalent to independent request-level serving under the specified frozen policy. The no-margin and certified conic policies have separate evolving feedback histories. The matched runner saves final policies, graphs, request laws, and per-window results locally for subsequent checking.

e) Repeated conditional sampling: For each data source we use the first test graph/split and freeze each of its 20 saved policies, separately for the no-margin and certified trajectories. Every policy is resampled 2,000 times with independent multinomial draws at its original window size. This gives 40,000 replications per method and data source. Pointwise one-sided 95% Clopper–Pearson intervals quantify conditional failure probability for each frozen plan. Simultaneous empirical inference over all saved plans would require multiplicity adjustment; Theorem M.2 instead supplies the adaptive trajectory guarantee under its sampling assumptions.

f) Margin sensitivity: On the first window's fixed score, take its conic optimum and search the witness segment for policies satisfying $g_i(\mathcal{Q}x) + c\Delta_i(x) \le -10^{-10}$, where $c \in \{0, 0.1, \dots, 1\}$ and $\Delta(x)$ is the final policy's analytic margin vector. The search uses 64 grid intervals and 35 bisection steps, checking every returned policy directly. The protocol uses 2,000 independent resamples at each point. At $c = 1$, the complete probability certificate holds. Smaller multipliers diagnose conservatism without the full theorem's failure budget. These tests diagnose model- and policy-specific conservatism; they do not establish zero failure probability or a universal margin reduction.

g) Data access: The public Last.fm archive is downloaded from its official source rather than bundled for redistribution.

S6. Diagnostic Simulation Protocols

These simulations isolate the two motivating mechanisms: reinforced global exposure can concentrate visibility, while within-user reinforcement can narrow individual menus. They test the scoped models of Section S2 and motivate CWOP's separate concentration and entropy constraints. Direct evaluation of CWOP's joint control is given in Section M.VII-D.

A. Supply Trajectory and Penalty Threshold

The supply simulation tests whether the regime ordering and corrective penalty remain visible when popularity and exposure are distinct states. The single-state identity $\beta_{\text{eff}} = \beta - \lambda/\tau$ is algebraic, so the simulation couples slow popularity f_c and fast exposure e_c through conversion q_{conv}. The single-state theory supplies a reference for interpreting the coupled simulation; the latter separates exposure from durable popularity.

a) Simulation protocol: The slow stock drives attachment through f_c^β, the fast flow carries the log-share penalty, and selection uses $w_c \propto f_c^\beta e_c^{-\lambda/\tau}$. Each served impression becomes durable popularity with probability q_{conv}. We use $K = 500$, $\tau = 1$, a Fenwick-tree sampler, and fixed seeds; the diagnostic contrasts use 30 to 50 seeds, while every cell in the horizon-penalty grid uses 60. We report HHI, top-1 share (the largest creator share), and the collapse fraction, defined as the fraction of seeds whose top-1 share reaches 0.5.

B. Per-User Exposure Entropy

The per-user simulation tests exposure narrowing under the same two-state dynamics. Here normalized entropy is $H(\sigma_u)/\log M$ for M options, and low entropy means this ratio is at most 0.5. The outcome is served exposure rather than consumption.

C. Simulation Details

The supply-side sweep uses horizons $T \in \{50\text{k}, 200\text{k}, 1\text{M}\}$, and penalty ratios $\lambda/\lambda^\star \in \{0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.65, 0.8, 1\}$. Each of the 27 cells uses 60 deterministic seeds. We save HHI, Gini (the normalized mean pairwise difference in exposure), top-1 share, and collapse indicators at top-1 thresholds 0.3, 0.5, and 0.7. Collapse fractions use Wilson 95% intervals. The aggregate state ratio $\sum_c f_c / \sum_c e_c$ measures conversion

accounting, not creator-level convergence. The separate linear-boundary supply run uses $\beta = 1.0$, $K = 500$, $T = 200{,}000$, and 50 seeds.

The per-user process uses $M = 50$ options and horizons $T \in \{5\text{k}, 20\text{k}, 100\text{k}\}$, matching the supply sweep's allocations per option. The super-linear grid uses the same nine penalty ratios and 60 seeds per cell. The linear-boundary comparison uses 100 seeds at $T = 100{,}000$ for $\lambda \in \{0, 0.05\}$.

a) Paired supply and demand protocol: The supply-side collapse and demand-side entropy sweeps use the same normalized penalty axis at $\beta = 1.35$, $\tau = 1$, and $q_{\text{conv}} = 0.2$. Each cell aggregates 60 fixed-seed runs. Horizons match allocations per option across sides ($T/K = T/M$); supply uncertainty uses Wilson 95% intervals, and demand uncertainty uses 95% intervals for the mean. The analytic threshold is the single-state reference for this controlled comparison.

D. Two-State Diagnostic Contrasts

The coupled simulator separates popularity from exposure. At $\beta = 1.35$ and $\lambda = 0$, it compares conversion probabilities $q_{\text{conv}} = 1$ and $q_{\text{conv}} = 0.2$ with a fixed-quality baseline, using 30 seeds per condition and $T = 200{,}000$. Seeds 0–29 differ from the 60-seed sweep starting at 2026072100. Cutoff checks use top-1 thresholds 0.3, 0.5, and 0.7 to assess whether the interpretation depends on a single collapse definition.

S7. Ethical Considerations and Compliance

Any future online experiment would require the applicable review and approval procedures for production A/B testing. It would be limited to content exposure and ranking mechanisms and would not alter content moderation policies, user privacy settings, or creators' control over content publication. The public Last.fm experiment uses only numeric artist-listening relations from the HetRec 2011 noncommercial release, not social links, names, or tags [1]. The reported formulation does not use sensitive attributes. Exposure constraints may redistribute creator opportunity and affect users' experience; they are not themselves a guarantee of fairness or improved well-being. Any production evaluation requires appropriate data-access permissions and institutional review, privacy safeguards, fairness monitoring, and an interference-aware design.

S8. Discussion

Joint concentration ceilings and per-user entropy floors allow CWOP to control exposure-based information oligopoly and narrow creator exposure within one allocation framework. The results make the associated utility cost explicit and show why finite-window serving requires safety margins in addition to feasible optimization.

S9. Extended Related Work

This section places CWOP within research on recommendation dynamics, fair allocation, and probabilistic constraints. These literatures contribute different parts of the problem: how recommendations change an ecosystem, which exposure distributions are desirable, and how reliably a randomized policy delivers those distributions.

a) Recommendation feedback and performativity: Chaney et al. [13] show through simulation that training on algorithmically confounded recommendation data can homogenize behavior without improving utility. Perdomo et al. [14] formalize the broader problem of performative prediction, in which deployment changes the distribution on which a predictor is evaluated, and analyze conditions for stable retraining. These results motivate treating recommendation as a sequential intervention rather than a prediction problem on a fixed dataset. CWOP addresses a different question within that sequence: conditional on the history available at the start of a window, can a selected exposure policy meet its targets during serving? Our certificate permits history-dependent policy updates between windows while requiring a known request law and conditionally independent sampling within each window. It establishes exposure safety under these assumptions, without requiring convergence to a performatively stable predictor.

b) Strategic creators and participation dynamics: Exposure changes creators' incentives as well as the data observed by a ranker. Eilat and Rosenfeld [15] design recommendations that anticipate strategic content creation and encourage content diversity. Jagadeesan et al. [16] characterize supply-side equilibria and producer specialization. Yao et al. [17] derive welfare guarantees for top-K recommendation under competing creators, showing that strategic feedback can have bounded welfare costs under a specified behavioral model. Participation introduces another mechanism: Mladenov et al. [18] formulate constrained matching to sustain viable providers, while Kiyohara et al. [19] study policy design when both sides' participation evolves. CWOP instead takes the current candidate structure and utility scores as inputs. Its safety margins protect served exposure as those scores change across windows. Predicting creator entry, content adaptation, or long-run ecosystem welfare would additionally require the behavioral and population models studied in this literature.

c) Exposure fairness and two-sided allocation: Singh and Joachims [20] optimize ranking utility subject to exposure-based fairness constraints, and Singh and Joachims [12] learn stochastic ranking policies with such constraints. Biega et al. [21] amortize individual fairness over a sequence of rankings, aligning accumulated attention with accumulated relevance. These works establish exposure allocation as a constrained decision problem. CWOP adopts that perspective but specifies aggregate HHI and maximum-share ceilings together with per-request-type entropy floors. These are structural diversity targets chosen by the platform; they do not by themselves define merit-based or demographic fairness. Do et al. [22] use Lorenz efficiency to improve the utilities of worse-off users and producers through concave welfare maximization. Our constrained utility objective and their distributional welfare objective encode different normative choices, so meeting CWOP's targets need not imply a Lorenz-efficient outcome.

d) Diversity, calibration, and the object being controlled: Wang and Joachims [23] show that user fairness, item fairness, and ranking diversity are distinct objectives that can conflict. Steck [24] calibrate recommendations to the proportions

of a user's past interests. Calibration preserves a reference distribution, whereas an entropy floor requires a minimum spread without selecting one particular distribution. CWOP measures this spread over creator identities in served exposure. Several creators can cover the same topic, and users can consume only a subset of what they receive. Thus creator exposure, topical diversity, and consumption diversity require separate measurements. This distinction determines both the interpretation of our entropy constraint and the scope of the experiments.

e) Fairness in expectation and in realized rankings: Randomization already plays a central role in fair ranking. Singh et al. [25] allocate ranking opportunities according to uncertainty about merit. Gorantla et al. [26] sample from distributions supported on rankings that satisfy group-representation constraints in every realization. CWOP's serving problem concerns a third object: nonlinear statistics of all exposures in a finite window, including the entropy within each request type. An expected exposure plan can satisfy these constraints while a sampled window exceeds them. Our margins bound that window-level risk. This complements ex-post ranking constructions: a sampler with dependent allocations or exact quotas could have different, potentially smaller fluctuations, but would require a guarantee for that sampling mechanism.

f) Online allocation and learning-based controllers: FairCo combines control of amortized exposure disparities with debiased relevance estimation in dynamic learning-to-rank [10]. FORM balances item and user fairness through constrained optimization and low-regret online learning [27]. Offr uses online Frank–Wolfe updates to optimize smooth concave objectives over rankings, exploiting their structure for efficient allocation [11]. These methods address learning or computational challenges beyond our implemented single-item setting. Our known-score adaptations isolate utility and constraint behavior under common serving conditions; their results apply to those adaptations. CWOP's certificate is independent of the optimizer that supplies a candidate policy. Accordingly, the conic reference and PD are alternative numerical implementations, and the independent reference measures the gap between target cost and solver error.

g) Chance constraints and safe sequential learning: Reserving slack to control random violations is a standard idea in stochastic optimization. Nemirovski and Shapiro [28] construct tractable convex approximations to chance-constrained programs. In constrained reinforcement learning, Liu et al. [29] obtain high-probability zero or bounded constraint violation using safe-policy assumptions and optimistic or pessimistic learning. CWOP conditions on the current request distribution and selected policy instead of estimating unknown transitions. Its certificate specializes constraint tightening to the joint behavior of share caps, HHI, and empirical entropy, accounting for random request-type counts and adaptive choices across windows. The fixed failure budget covers the prescribed collection of windows. Extending that guarantee to an unknown request law, misspecified response estimates, or unrestricted within-window adaptation requires additional uncertainty bounds.

h) Predictive control and adaptive replanning: Stochastic MPC repeatedly optimizes a sequence of controls using a dynamical model, with chance constraints or deterministic tightenings to account for uncertainty [30]. CWOP shares the use of tightened constraints and repeated optimization. Its current-window score is supplied by the observed history rather than a model predicting future states. The serving theorem conditions on that history and protects sampled exposure over a prescribed sequence of windows. Its specific contribution is the joint analysis of cap, HHI, and per-type empirical-entropy statistics, including random request-type counts and a computable utility budget. Recursive feasibility after changing candidates or request laws, closed-loop stability, and look-ahead utility concern additional dynamical properties beyond this window-level exposure guarantee.

i) Convex optimization and the cost of constraints: The request-policy formulation uses standard convex analysis: linear expected utility, convex concentration ceilings, and superlevel sets of concave entropy. Boyd and Vandenberghe [31] provide the foundations for duality, Slater conditions, and barrier methods. Nedić and Ozdaglar [6] analyze primal averaging and infeasibility bounds for dual subgradient methods. These results support our numerical convergence analysis and multiplier certificates. The feasible witness also supplies a concrete way to repair an approximate policy and reserve additional slack. Our cost decomposition separates the loss imposed by the original exposure targets from numerical error and the further loss needed for random serving. The uniform-envelope argument connects this mixture construction to policy-dependent certification, giving a utility lower bound even though a policy-dependent certified set need not be convex.

j) Concentration, entropy, and reinforcement foundations: Variance-sensitive and bounded-difference inequalities are classical tools for functions of independent random variables [32]. Entropy, relative entropy, and the method of types are developed systematically by Cover and Thomas [33]; finite-sample entropy estimation and bias are studied by Paninski [8]. We combine these tools using the variance of a proposed exposure policy and the decomposition of plug-in entropy error into a linear term and relative entropy. Conditioning at the start of each window preserves the required sampling assumptions. The local cap-margin necessity result uses classical central limit reasoning [34] and establishes the $N^{-1/2}$ order for a nondegenerate cap boundary, rather than tightness of every margin constant. Separately, Pemantle [3] surveys reinforced processes underlying the motivating urn examples. Those examples describe path dependence; the serving theorem applies without identifying a reinforcement exponent.

k) Identification and empirical interpretation: Schnabel et al. [35] develop methods for learning and evaluation under selection-biased recommendation data. Their perspective explains why descriptive production concentration cannot identify the effect of deploying a new exposure policy. Controlled

feedback experiments and Last.fm-derived preference proxies test optimization and sampling safety. Taken together, the contribution is a quantified connection between a specified exposure plan, its utility cost, and finite-window delivery.

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